

GENERICS

// class
class Box<T> { }
// variable
Box<String> b = new Box<String>();
// method
public <T> void doStuff(T value) { }

JAVA BEING STUPID

T obj = new T(); // error
obj instanceof T; // error
T heh = (T) "asdf"; // no typechecking
var s = Stack<int>(); // no primitive types
private static T x; // static no worky
Node<Number> s = new Node<Integer>; // error
class IThrowAnErrorBecauseWhyNot {
 void m(List<String> list) { }
 void m(List<Integer> list) { }
} // error because of type erasure & identifiability
Integer[] c = set.toArray(new Integer[0]);

ITERATOR

for (String s : stringList) { // ==
 for (Iterator<String> i = stringList.iterator();
 i.hasNext();) {
 String s = i.next();
 }
}

ITERABLE

interface Iterable<T> {
 Iterator<T> iterator();
}
interface Iterator<E> {
 boolean hasNext();
 E next();
}

COMPARABLE

TODO:

Type signature?

static <T extends Comparable> T doStuff(T a, T b) {
 // compareTo is possible with all types
 return a.compareTo("b") > 0 ? a : b;
}
static <T extends Comparable<T>> T doStuff(T a, T b)
{
 // compareTo is possible only with T
 return a.compareTo(b) > 0 ? a : b;
}

EXAMPLE

class Graphic {
 public void draw() { }
}
class Rectangle extends Graphic { }
class Circle extends Graphic { }
class GraphicStack<T extends Graphic>
 extends Stack<T> {
 public void drawAll() {
 for (T item : this) item.draw();
 }
}

BYTECODE

TODO:

TYPE ERASURE

– Introduced because of "bAcKwArDs CoMpAtAbIlItY"
– No type information at runtime (Non-Reifiable Type)
class MyStack<T> {
 Entry<T> top;
 int push(T value) { }
}
// becomes
class MyStack {
 Entry top;
 int push(Object value) { }
}
/* with bounds */
class MyStack<T extends Comparable<T>> {
 Entry<T> top;
 int push(T value) { }
}
// becomes

class MyStack {
 Entry top;
 int push(Comparable value) { }
}
/* what happens at/before runtime */
MyStack<String> s = new MyStack<String>();
s.push("Eh");
String x = s.pop();
// becomes
MyStack s = new MyStack();
s.push("Eh");
String x = (String) s.pop();

WILDCARDS

void printGs(List<Graphic> gs) { }
List<Graphic> gs1 = new ArrayList<>();
printGs(gs1); // ok
List<Circle> gs2 = new ArrayList<>();
printGs(gs2); // error
/* solution */
void printGs(List<? extends Graphic> gs) { }

GENERIC VARIANCE

	Type	Compatible Type-Args	R	W
Invariance	C<T>	T	✓	✓
Covariance	C<? extends T>	T and sub-types	✓	×
Contravariance	C<? super T>	T and basis-types	×	✓
Bivariance	C<?>	All	×	×

INVARIANCE

static <T> void move(Stack<T>, from, Stack<T> to) {
 while(!from.isEmpty()) to.push(from.pop());
}
var rectS = new Stack<Rectangle>();
var graphS = new Stack<Graphic>();
graphS.push(new Rectangle()); // ok
move(rectS, graphS); // error
Stack<Object> objS = graphS; // error
/* anotherone */
String[] strA = new String[10];
Object[] objA = strA; // ok
objA[0] = Integer.valueOf(2); // runtime err

TODO:

// compiles?
List<Graphic> list = new ArrayList<Rectangle>();
// compiles?
Rectangle[] rectangleStack = new Rectangle[10];
Graphic[] graphicStack = new Graphic[10];
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
graphicStack = rectangleStack;
for (Rectangle r : rectangleStack) {
 graphicStack.push(r);
}

COVARIANCE

public class Stack<E> {
 public void pushAll(Iterable<? extends E> src) { }
}
Stack<? extends Graphic> graphS;
graphS = new Stack<Rectangle>(); // ok
graphS = new Stack<Circle>(); // ok
graphS = new Stack<Object>(); // error (duh)
/* read only */
Stack<? extends Graphic> stack = new
 Stack<Rectangle>();
stack.push(new Graphic()); // error
stack.push(new Rectangle()); // error
stack.push(new Triangle()); // error

CONTRAVARIANCE

static <T> void addToC(List<? super T> list, T e) {
 list.add(e);
}
addToC(new ArrayList<Number>(), 3); // ok
addToC(new ArrayList<Object>(), 3); // ok

/* shenanigans */
Stack<? super Graphic> s = new Stack<Object>();
s.add(new Graphic()); // ok
s.add(new Rectangle()); // ok
Graphic g = s.pop(); // error
Object g = s.pop(); // ok

BIVARIANCE

static void printList(List<?> list) {
 for (Object elem : list)
 System.out.print(elem + " ");
}
/* more shenanigans */
static void appendNewObject(List<?> list) {
 list.add(new Object()); // error
}

PRODUCER / CONSUMER

TODO:

MISC

<T extends Comparable & Collection> // multiple type
 bounds

GENERICS STREAM TOMFOOLERY

List<? extends Media> medialis;
public <S extends T> List<S>
 filterBySubtype(Class<S> subtype) {
 return medialis.stream()
 .filter(subtype::isInstance)
 .map(subtype::cast)
 .collect(Collectors.toList());
 }
}

ANNOTATIONS

– Metadata
– Not actually part of the code

TODO:

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– @Override
– @Test
– @Json...

DEFINING

@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface Author {
 String name();
 String date() default "N/A";
}

@Author(name = "UwU", date = "idon't validate input")
public class WeeDoo { }

REFLECTON

– Information of Class (private and public): Methods, Attrib-
 utes, Parent class, Implemented interfaces
– Calling methods possible
– Usages: Debugger, Serialization, Frameworks, Remote
 Procedure Call, ORM
// all of these `throws SecurityException`
public Field[] getDeclaredFields()
public Method[] getDeclaredMethods()
public Constructor<?>[] getDeclaredConstructors()
System.out.println(dump(new Student("a", "b"), 0));
// {
// firstName = a
// lastName = b
// }
Class c = "s".getClass(); // java.lang.String

TODO:

Diagram (slides 26)

METHOD

@Target(ElementType.METHOD)

public String getName()
public boolean isAnnotationPresent(
 Class<? extends Annotation> annotationClass)
public Object invoke(Object obj, Object... args)

throws IllegalAccessException,
 IllegalArgumentException,
 InvocationTargetException

public class Profiler {
 public static void main(String[] args) {
 var testFunctions = new ProfileFunctions();
 int[] array = {5, 2, 8, 1, 93, 3, 33, 1, 333};
 var methods = ProfileFunctions.class
 .getDeclaredMethods();
 for (var m : methods) {
 if(m.isAnnotationPresent(Profile.class)) {
 m.setAccessible(true); // best practice or
 sth, idk, i write code in actually good languages
 Profiler.profileMethod(testFunctions, m,
 new Object[] {array});
 }
 }
 }
}

CLASS

TODO:

@Target(ElementType.TYPE)

getDeclaredConstructor().newInstance()
...
ATTRIBUTE

TODO:

@Target(ElementType.FIELD)

...
VALIDATION

@Min(value = 18, message = "Age must be ≥ 18")
@Max(value = 99, message = "Age must be ≤ 99")
private int age;
@NotNull(message = "Name cannot be null")
private String name;

ALGORITHMS

– Brute-force
– Greedy
 – Take best option each step (optimize locally)
 – Fast
– Backtracking
 – Trial and error
 – Best overall option
 – Higher compute costs
– Divide and conquer
 – Binary search
– Dynamic programming
 – Recursion optimization?
 – Tabulation

BACKTRACKING

TODO:

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BIG O NOTATION

– Worst case scenario
– Atomic operations = constant time
– Runtime measured as sum of primitive operations

Notation

$$\widetilde{f} \left(\underbrace{n}_{\text{Size of problem}} \right) = \text{Number of steps}$$

g = complexity class

TODO:

diagram $f \leq cg$

$f(n)$ is $O(g(n))$ if $c > 0 \wedge n_0 \geq 1$
 $f(n) \leq cg(n)$ for $n \geq n_0$

f in order g:

Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$
 $n_0, c \in \mathbb{N} \wedge \forall n \geq n_0 : f(n) \leq c \cdot g(n)$
 $\Rightarrow f \in O(g) \Leftrightarrow f = O(g)$

Big O

$$\widetilde{O} \left(\underbrace{n}_{\text{Size of problem}} \right) = \text{Complexity class}$$

RULES

If $f(n)$ is polynomial of degree d , then $f(n) \in O(n^d)$
– ignore lower powers
– ignore constants
Use most optimal function (lowest power)
– $2n$ is $O(n)$ better than $2n$ is $O(n^2)$

Simplify as far as possible
– $3n + 5$ is $O(n)$ instead of $3n + 5$ is $O(3n)$

COMPLEXITY CLASSES

Name	Class	Example
Constant	1	Array read
Logarithmic	$\log n$	Binary search
Linear	n	Linear search
Log-linear	$n \log n$	Merge+Quick sort
Quadratic	n^2	Bubble+Insertion sort
Cubic	n^3	Solve equation
Polynomial	n^k	Various algorithms
Exponential	2^n	Calculate all subsets
Factorial	$n!$	Calculate all permutations

$O(1)$
 $O(\log n)$
 $O(n)$
 $O(n \log n)$
 $O(n^2)$
 $O(n^3)$
 $O(2^n)$

TODO:

Counting primitive operations (slides 55)

SORTING ALGORITHMS

SELECTION SORT

– Search for smallest elem in unsorted part and move in front

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